Unit 2 Guided Notes — Exponents & Scientific Notation

Same base? Move the exponents. Big or tiny number? Pack it into a × 10^n.

Standards: NC.8.EE.1 · NC.8.EE.3 · NC.8.EE.4

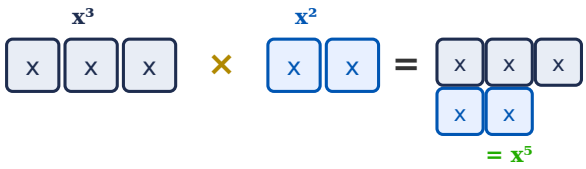
Blank Worksheet (PDF) | ✓ Answer Key (PDF) | Official PDF Packet | ♿ Version 2 (PDF)

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Work the blanks on your printed copy first, then click any blank, step, or check button on screen to reveal the answer.

1 Exponent Rules

📌 **KEY RULE:** Exponent rules only apply when the BASES are the SAME!

Product Rule — Count the Factors



: $x^2 = x^{3+2} = x^5$ → count the factors: 3 + 2 = 5 → ADD the expone

The Five Core Rules

Fill the examples and key phrase on your printed copy, then click to check.

Rule	Formula	Simple Example	Monomial Example	Key Phrase
Product Rule	$x^a \times x^b = x^{a+b}$	$x^3 \cdot x^4 = x^7$	$(2x^3)(3x^2) = 6x^5$	Same base? ADD
Quotient Rule	$x^a \div x^b = x^{a-b}$	$x^6 \div x^2 = x^4$	$(12x^7) \div (3x^2) = 4x^5$	Same base? SUBTRACT
Power Rule	$(x^a)^b = x^{axb}$	$(x^2)^3 = x^6$	$(2x^3)^2 = 4x^6$	Power to power? MULTIPLY
Zero Exponent	$x^0 = 1$	$5^0 = 1$	$(3x^5)^0 = 1$	ANY base^0=1
Negative Exponent	$x^{-n} = 1/x^n$	$2^{-3} = 1/8$	$4x^{-3} = 4/x^3$	Flip to fraction

⚠ **PARENTHESES MATTER!** -3^2 = **−9** vs. $(-3)^2$ = **9**. Without ( ), ONLY the base gets the exponent — not the negative sign!

Modeling the Rules — See WHY

Each rule comes from writing out the repeated multiplication. Click each card to reveal the reasoning.

Product Rule · $x^3 \times x^2 = x^5$

$x \cdot x \cdot x \times x \cdot x = x \cdot x \cdot x \cdot x \cdot x = x^5$. Count the factors: 3 + 2 = 5. Same base → ADD the exponents.

Quotient Rule · $x^5 \div x^2 = x^3$

$x \cdot x \cdot x \cdot x \cdot x \div x \cdot x \rightarrow$ cancel 2 matching pairs → $x \cdot x \cdot x = x^3$. Cancel matching factors: 5 − 2 = 3. Same base → SUBTRACT the exponents.

Power Rule · $(x^2)^3 = x^6$

Three copies of x^2: $(x \cdot x)(x \cdot x)(x \cdot x) = x^6$. 2 × 3 = 6. Power to a power → MULTIPLY.

Zero Exponent · $x^0 = 1$

$x \cdot x \cdot x \div x \cdot x \cdot x = 1$ (anything divided by itself is 1) — but by the quotient rule this also equals $x^{3-3} = x^0$. So $x^0 = 1$.

Negative Exponent · $x^{-3} = 1/x^3$

$x \cdot x \div x \cdot x \cdot x \cdot x = x^{2-5} = x^{-3}$ — but cancelling factors leaves $1/x^3$. So a negative exponent means "flip it": $x^{-n} = 1/x^n$.

Quick Check

MC **Simplify $x^5 \cdot x^3$. Which rule applies, and what is the result?**

✓ Same base, multiplication → Product Rule: ADD the exponents. $x^5 \cdot x^3 = x^8$.

Match the Rule to its Key Phrase

MATCH **Click a rule, then its key phrase.**

Rule

- Product Rule
- Quotient Rule
- Power Rule
- Negative Exponent

Key Phrase

- Power to power? MULTIPLY
- Same base? ADD exponents
- Flip to a fraction

Same base? SUBTRACT exponents

0 of 4 matched

Worked Example — Simplify $(4x^3y^2)(3x^5y)$

Step 1 — Multiply coefficients: $4 \times 3 = 12$

Step 2 — Add x exponents: $x^3 \cdot x^5 = x^8$

Step 3 — Add y exponents: $y^2 \cdot y^1 = y^3$

✓ **Step 4 — Combine:** $12x^8y^3$

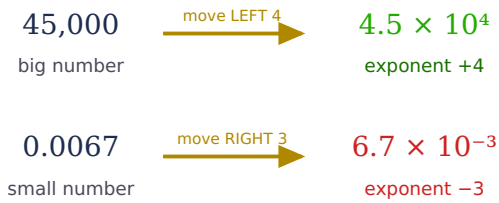
ERROR ANALYSIS **Simplify $(3x^2)^3$. A student wrote $3x^6$. What went wrong?**

The coefficient 3 must ALSO be cubed: $3^3 = 27$. Correct: $27x^6$. The power rule applies to EVERYTHING inside the parentheses.

2 Scientific Notation

📌 **THE FORMAT:** $a \times 10^n$, where $1 \leq a < 10$ — example: 4.5×10^7 = coefficient · base (always 10) · exponent.

Move the Decimal



Type	Direction	Exponent	Example
Large number	Move decimal LEFT	POSITIVE	$45{,}000 = 4.5 \times 10^4$
Small number (< 1)	Move decimal RIGHT	NEGATIVE	$0.0067 = 6.7 \times 10^{-3}$

Multiplying — 3-Step Process

1. MULTIPLY the coefficients (the a-values)
2. ADD the exponents
3. Adjust if coefficient is NOT between 1 and 10

Example (no adjustment): $(3 \times 10^4) \times (2 \times 10^5)$

Multiply coefficients: $3 \times 2 = 6$

ADD exponents: 4 + 5 = 9

✓ **Answer: 6×10^9**

Dividing — 3-Step Process

1. **DIVIDE** the coefficients
2. **SUBTRACT** the exponents
3. Adjust if needed

Example (no adjustment): $(8 \times 10^6) \div (4 \times 10^2)$

Divide coefficients: 8 ÷ 4 = 2

SUBTRACT exponents: 6 − 2 = 4

✓ **Answer: 2×10^4**

⚠ **ADJUSTMENT RULES:** If coefficient ≥ 10: move decimal LEFT, ADD 1 to exponent. If coefficient < 1: move decimal RIGHT, SUBTRACT 1 from exponent.

Example needing adjustment: $(5 \times 10^3) \times (4 \times 10^2)$

Multiply: 5 × 4 = 20

Add exponents: 3 + 2 = 5 → result: 20×10^5

Adjust (20 ≥ 10): $20 = 2.0 \times 10^1$, so ADD 1 to the exponent

✓ **Final answer: 2.0×10^6**

ERROR ANALYSIS **Convert 0.0045 to scientific notation. A student wrote 4.5×10^3. What's the mistake?**

Small number → NEGATIVE exponent. Correct: 4.5×10^{-3}. Numbers less than 1 always have negative exponents in scientific notation.

✏ Go to Unit 2 Practice →

✏ Part 2 — Practice Problems

📌 **Show all work on your printed copy.** Circle your final answer, then check on screen.

A. Exponent Rules with Monomials

1) Simplify $x^3 \cdot x^5$

answer

2) Simplify $(y^4)^3$

answer

3) Simplify $a^8 \div a^3$

answer

4) Simplify $(2x^3)(5x^2)$

answer

5) Simplify $(3m^2)^3$

answer

6) Simplify $12y^7 \div 4y^2$

answer

7) Simplify $(-2a^4)^2$

answer

8) Simplify $(5x^2y)(3xy^3)$

answer

9) Simplify $18n^6 \div 6n^6$

answer

10) Simplify $(4b^3)^2 \div 2b$

answer

11) Simplify $(-3x^2)(-4x^3)$

answer

12) Simplify $(x^5y^2)^3$

answer

B. Scientific Notation Conversions

1) 45,000,000 → scientific notation

answer

2) 0.00032 → scientific notation

answer

3) 6.02×10^8 → standard form

answer

4) 9.1×10^{-5} → standard form

answer

5) 780,000 → scientific notation

answer

6) 0.0000067 → scientific notation

answer

7) 3.45×10^4 → standard form

answer

8) 1.2×10^{-3} → standard form

answer

9) 5,280,000,000 → scientific notation

answer

10) 2.5×10^{-2} → standard form

answer

C. Operations

1) $(3 \times 10^{4})(2 \times 10^{5})$

answer

2) $(8 \times 10^{7}) \div (4 \times 10^{3})$

answer

3) $(5 \times 10^{3})(7 \times 10^{6})$

answer

4) $(9.6 \times 10^{8}) \div (3.2 \times 10^{5})$

answer

D. Word Problems

1. The Sun is about 9.3×10^{7} miles from Earth. Light travels 1.86×10^{5} miles per second. How many seconds does it take light to reach Earth?

$(9.3 \times 10^{7}) \div (1.86 \times 10^{5}) = 5 \times 10^{2}$ seconds (500 seconds).

2. A bacterium is about 2×10^{-6} m long. A virus is about 5×10^{-8} m long. How many times longer is the bacterium?

$(2 \times 10^{-6}) \div (5 \times 10^{-8}) = 0.4 \times 10^{2} = 40$ times longer.

3. The US national debt is about 3.4×10^{13} dollars. The US population is about 3.4×10^{8} people. What is the debt per person?

$(3.4 \times 10^{13}) \div (3.4 \times 10^{8}) = 1 \times 10^{5} = \$100{,}000$ per person.

4. A grain of sand has mass 6.5×10^{-4} grams. A beach has about 8×10^{18} grains of sand. What is the total mass in grams?

$(6.5 \times 10^{-4})(8 \times 10^{18}) = 52 \times 10^{14} = 5.2 \times 10^{15}$ grams.

5. There are about 3.7×10^{13} cells in a human body. Each cell is about 1×10^{-5} m wide. If all cells were lined up, how long in meters?

$(3.7 \times 10^{13})(1 \times 10^{-5}) = 3.7 \times 10^{8}$ meters.

6. A computer does 4.8×10^{9} calculations per second. A task needs 1.44×10^{12} calculations. How many seconds to finish?

$(1.44 \times 10^{12}) \div (4.8 \times 10^{9}) = 0.3 \times 10^{3} = 3 \times 10^{2}$ seconds (300 seconds).

← Back to Unit 2